Workflow and visualization of age populations:
(Method presented in *Synchronized cyclicity in a magmatic arc and back-arc over 80 Ma*)

The following scripts can be executed on any standard computer with Python 3 and the required libraries (numpy, pandas, openpyxl). No specific operating system is required. Parallel processing is implemented via ProcessPoolExecutor, and the number of cores used can be controlled by the user (N_CORES in S1_EM_bootstrapp_.py). Performance depends primarily on the number of cores, the number of bootstrap iterations (N_BOOTSTRAP) and the dataset size.

Step 0: building Excel files (.xlsx)

A file in the format exemplified in "age_distributions_input.xlsx" is required. It contains the list of distributions and their IDs (including the four sample sizes). The format includes alternating rows, where the first row contains the sample ID (e.g., Age Distribution 1) and ages, and the subsequent row contains the corresponding two-sigma uncertainties (Fig. 1):

	A	B	C	D	E	F	G
1	Age Distribution 1	462.0	426.7	455.7	430.1	469.2	458.0
2	2σ	5.6	8.6	4.1	5.0	5.3	5.1
3	Age Distribution 2	421.0	445.6	433.1	448.5	431.8	465.5
4	2σ	14.5	15.0	15.0	15.0	15.0	18.8
5	Age Distribution 3	459.8	459.0	462.1	473.8	494.1	462.0
6	2σ	15.2	15.9	16.2	17.4	19.7	15.7
7	Age Distribution 4	457.2	461.2	470.1	467.7	481.3	484.4
8	2σ	16.7	15.6	16.0	17.4	17.2	17.8

Fig. 1

You must create an Excel spreadsheet like this for: (1) your observed/unknown distributions (one file); and (2) for your training synthetic distributions. For the latter, you will need to create a file for each mixture scenario you want to test. For example, if you want to test four mixture scenarios, where the number of mixed populations range from N = 2 to N = 5, you will need an Excel file for each N-mixture distribution (N = 2, N = 3, N = 4, and N = 5), for a total of four files.
If there are multiple samples from the same distribution, list them in rows in the format shown in Fig. 1, and specify the sample size in brackets. For example, if there are samples of size 800, 120, 60, and 40, list them as follows:

	A	B	C	D	E	F	G
1	Synthetic Age Distribution 1 (800)	462.0	426.7	455.7	430.1	469.2	458.0
2	2σ	5.6	8.6	4.1	5.0	5.3	5.1
3	Synthetic Age Distribution 1 (120)	462.0	494.1	462.0	430.1	462.0	426.7
4	2σ	5.6	12.0	15.0	9.0	8.5	7.0
5	Synthetic Age Distribution 1 (60)	457.9	459.9	430.1	469.2	453.5	465.5
6	2σ	10.3	15.0	12.0	15.0	10.3	18.8
7	Synthetic Age Distribution 1 (40)	457.2	465.7	475.0	467.7	481.3	465.5
8	2σ	13.0	15.6	16.0	10.0	6.2	18.8
9	Synthetic Age Distribution 2 (800)	457.9	459.9	465.5	447.7	453.5	465.5
10	2σ	5.6	15.0	15.0	15.0	15.0	18.8
11	Synthetic Age Distribution 2 (120)	467.4	466.1	462.1	473.8	494.1	462.0
12	2σ	15.2	10.0	9.5	8.5	19.7	15.7
13	Synthetic Age Distribution 2 (60)	457.2	465.7	475.0	467.7	481.3	484.4
14	2σ	16.7	15.6	16.0	17.4	17.2	17.8
15	Synthetic Age Distribution 2 (40)	457.2	465.7	475.0	467.7	481.3	484.4
16	2σ	7.0	10.0	15.0	15.0	13.0	17.8

Fig. 2.

Therefore, a spreadsheet like the one shown in Fig. 2 will be needed for each N-mixture.

STEP 1: run **S1_EM_bootstrapp_.py**

This script implements a heteroscedastic Gaussian mixture model (GMM) using an expectation-maximization (EM) algorithm based on the work of Sambridge and Compston (1994), combined with bootstrap validation. The script is designed to deconvolve complex geochronological age distributions into statistically distinct populations, assuming four mixing scenarios (N = 2 to N = 5). For all age distributions (and age uncertainties) in the age_distributions_input.xlsx files, the S1_EM_bootstrapp_.py program deconvolves each distribution (each row) in four (by default N = 2 to 5) possible mixing scenarios and assigns individual ages to specific populations with calculated confidence levels. Additionally, for each scenario and for each analyzed distribution, the code runs the EM algorithm 500 times (by default) to obtain multiple results from the same scenario. This provides uncertainty estimates for the parameters of the deconvoluted populations (bootstrapping).

Required Input Files:

1. N-files of the type age_distributions_input.xlsx explained in Step 0.

Generated Output Files:

Each time S1_EM_bootstrapp_.py is run, three (3) output files are returned that will be necessary for the next scripts:

1. Validation_Results_.xlsx: Contains the BIC mean normalized to the sample size (n) and the detected population parameters (population means in columns "PopN_Mean" and their uncertainties via bootstrapping in columns "PopN_SD" and "PopN_Prop") for each assumed N (called "k" in the file). Please note that the "PopN_SD" corresponds to the mean uncertainty calculated via 500 repetitions, **not the standard deviation of the**

population. Also, **in this spreadsheet, N represents the number of analyses in the population. Do not confuse it with N-mixture ("k" here) (my mistake! sorry).**

2. BF_DIFFS_.xlsx: Detailed Bayes Factors (BF) for each model transition (e.g., N=2 to N=3), used to statistically justify the selection of the optimal number of populations.

3. Grain_Assignments_.xlsx: An age-by-age breakdown showing the specific population assignment and the percentage of confidence for each individual age of pertaining to the selected population.

Important: S1_EM_bootstrap.py produces three files with the same name for each run. Therefore, it is important to rename or move the files from each run to different folders. Otherwise, S1_EM_bootstrap.py will overwrite the previous run's files, and they will have to be run again.

Python Dependencies:

- numpy: Numerical computing and vectorized matrix operations.
- pandas: Data manipulation and Excel I/O.
- openpyxl: Engine for reading/writing .xlsx files.
- concurrent.futures: Parallel processing to accelerate Bootstrap simulations.

STEP 2: run S2_N_selection.py

This script selects the most probable number of populations (N) of the observed dataset based on your synthetic distributions' dataset using the proposed multivariate fingerprinting.

Before executing this script, rename the files returned by S1_EM_bootstrap_.py from BF_DIFFS_.xlsx to the following: For example, for four mixture scenarios, there will be five files: BF_DIFFS_Obs_.xlsx, BF_DIFFS_N2.xlsx, BF_DIFFS_N3.xlsx, BF_DIFFS_N4.xlsx, and BF_DIFFS_N5.xlsx. This is the only naming convention recognized by the program.

Required Input Files:

Since the script runs both the estimation and validation functions simultaneously, all five input files for the default scenario of N = 2 to 5 must be present in the working directory to avoid execution errors.

All inputs are derived from the outputs of S1_EM_bootstrapp_.py:

1. Training Set (BF_DIFFS_N2.xlsx through BF_DIFFS_N5.xlsx): These files contain the Bayes factors for known scenarios where the number of populations is pre-defined.

2. Testing Set (BF_DIFFS_Obs_.xlsx): This is the file containing the samples you wish to test. It is renamed specifically to distinguish it from the training library.

It is essential to place the sample size in parentheses in each 'BF_DIFFS_' file distribution (Fig. 3):

	A	B	C	D
1	Sample	BF_2to3_n	BF_3to4_n	BF_4to5_n
2	age1 (800)	1.127691	0.991524	1.001783
3	age1 (120)	1.026428	0.980105	0.955782
4	age1 (60)	1.0444	0.951905	0.939458
5	age1 (40)	1.088296	0.92124	0.917798
6	age2 (800)	1.163799	0.996929	0.993063
7	age2 (120)	1.047456	1.01081	0.959311
8	age2 (60)	0.947973	0.968436	0.947226
9	age2 (40)	0.916436	0.92211	0.906616
10	age3 (800)	1.041661	1.003046	0.992794

Fig. 3

The script is configured for these four sample sizes of the article (800, 120, 60, and 40), but it can easily be modified (see the line of the script in blue in Fig. 4):

```
# BLOQUE 2: CONFIGURACION
#

PATH = '/Users/marcosmorfulis/Desktop/' ###Directorio
ESC = [2, 3, 4, 5]
TAMS = [800, 120, 60, 40]

df_master, mi_biblioteca, dim_final = ejecutar_sistema
```

Fig. 4.

Generated Output Files:

1. Classification_Results.xlsx: Final classification of the "Observed", real-world samples, including Confidence % and Log-Likelihood.
2. Confusion_Matrices_.png: Heatmaps showing the method's success rate in identifying known scenarios.

3. Confusion_Matrix_abs_.xlsx: Raw counts for the validation phase, separated by sample size (n). |

Python Dependencies:

- numpy & pandas: Data handling and log-transformations.
- scipy.stats (multivariate_normal): The core probabilistic classification engine.
- sklearn: For confusion_matrix (validation), and train_test_split.
- matplotlib & seaborn: For "publication-ready" graphical outputs.

STEP 3 (optional): run S3_Statist_pops.py

Simply run the script after setting the files path. Although optional, this is recommended in order to ascertain whether the selected N-mixture model and its deconvolved populations are statistically sound. Ideally, a MSWD close to 1 should be recovered for each population after deconvolution. The Lilliefors test for normality also helps to test the assumption of Gaussian populations in the mixture.

Required Input Files:

1. Grain_Assignments_.xlsx
2. Classification_Results.xlsx

Generated Output Files:

1. Statistical_Diag_.xlsx: contains the following columns:

   - Sample: sample name (Id)
   - PopL Population label (P1, P2, …)
   - N: Number of analyses in the population. **Caution!** Do not confound with N-mixture (my mistake! … again)
   - MSWD: Dispersion relative to analytical uncertainties
   - Lilliefors_p: value of the normality test
   - Normal? "Yes" if Lilliefors $p > 0.05$, otherwise "No"

STEP 4 (optional): run S4_gmm_plot.py

Simply run the script after setting the sample name and file path. This script generates a histogram of grain ages for a selected sample or distribution, along with Gaussian population curves derived from a mixture model. It combines the results of previous classifications, validations, and statistical diagnostics to produce a final interpretive plot. The uncertainty bands are shown as gray shaded regions around each population mean. They represent $\pm 2\sigma$ of the bootstrap-based, providing an estimate of the uncertainty in the population mean.

Required Input Files:

The script requires four Excel files**:**

1. Validation_Results_.xlsx
2. Grain_Assignments_.xlsx
3. Classification_Results.xlsx
4. Statistical_Diag_.xlsx